Supplementary material for "Thermodynamics of CO_2-loaded aqueous amines with ePC-SAFT. Part II: transfer to MDEA, AMP, piperazine and DEA"

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S1 Shared and amine-specific inputs

Main-text Section 2 defines the equation of state, combining rules, relative permittivity, Born term, reaction conditions and caloric functions; this section lists the shared scalar inputs and the reaction-law coefficients.

The complete electronic files also retain temperature-dependent coefficients, association-site multiplicities, correlations and the parameter ties. The electronic parameter files specify the neutral donor–acceptor multiplicities, the charged MDEA and DEA chains, the single-sphere AMP and PZ ions and carbamates, and the nonassociating HPZCOO zwitterion. Shared component and pair values remain fixed; an interaction between a new amine species and a shared component can be fitted without changing the shared component itself. No fitted ionic size is interpreted as an independent molecular measurement.

The shared Born diameters and the water interactions other than the fitted bicarbonate term are assumed fixed inputs transferred from the MEA parameter set [1], rather than inorganic-ion size measurements. The bicarbonate–water interaction is fixed at the MEA fit's upper bound. The MEA joint-fit R2 coefficients retain their conditional uncertainty in Part I, but no uncertainty of those shared inputs is propagated here.

Table S1
Shared species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
CO_2	ϵ/k_B	169.21	K
CO_2	m^{seg}	2.0729	1
CO_2	σ	2.7852	Å
CO_2	f	1	1
CO_2	d_V	1×10^{-8}	L/mol
CO_2	d_T	1403	K
water	ϵ/k_B	432.37881	K
water	m^{seg}	1.2527538	1
water	f	1.5	1
water	d_V	0.3777	L/mol
water	d_T	1403	K
HCO_3^-	d^{Born}	3	Å
HCO_3^-	d^{DH}	2.9296	Å
HCO_3^-	ϵ/k_B	70	K
HCO_3^-	d^{pack}	2.578048	Å
HCO_3^-	m^{seg}	1	1
HCO_3^-	σ	2.9296	Å
HCO_3^-	f	1	1
CO_3^{2-}	d^{Born}	3	Å

Species	Quantity	Value	Unit
CO_3^{2-}	d^{DH}	2.4422	Å
CO_3^{2-}	ϵ/k_B	249.26	K
CO_3^{2-}	d^{pack}	2.149136	Å
CO_3^{2-}	m^{seg}	1	1
CO_3^{2-}	σ	2.4422	Å
CO_3^{2-}	f	1	1
H_3O^+	d^{Born}	1.218	Å
H_3O^+	d^{DH}	3.4654	Å
H_3O^+	ϵ/k_B	500	K
H_3O^+	d^{pack}	3.049552	Å
H_3O^+	m^{seg}	1	1
H_3O^+	σ	3.4654	Å
H_3O^+	f	1	1
OH^-	d^{Born}	3.0810769	Å
OH^-	d^{DH}	2.0177	Å
OH^-	ϵ/k_B	650	K
OH^-	d^{pack}	1.775576	Å
OH^-	m^{seg}	1	1
OH^-	σ	2.0177	Å
OH^-	f	1	1

Table S2
Shared nonzero constant pair coefficients and association parameters. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
CO_2–water	k_{ij}	−0.0464169
CO_2–water	b_{ij} / K	−35.6650522
HCO_3^-–water	k_{ij}	1.0000000
CO_3^{2-}–water	k_{ij}	−0.2500000
H_3O^+–water	k_{ij}	0.2500000
OH^-–water	k_{ij}	−0.2500000
water–water	ϵ^{AB}/k_B / K	1756.6497965
water–water	κ^{AB}	0.0376197
CO_2–water	ϵ^{AB}/k_B / K	878.3248982
CO_2–water	κ^{AB}	0.0376197
CO_2–water	ϵ^{AB}/k_B / K	878.3248982
CO_2–water	κ^{AB}	0.0376197

S1.1 Reaction directions, coefficients and reference states

The R1 and R3 source constants have a mole-fraction infinite-dilution basis, while R2 and the amine laws have a common-molality basis. The tables retain those source coefficients rather than silently substituting a different basis.

The dissociation sources are Kamps and Maurer for MDEA, Hamborg et al. for AMP and PZ protonation, Ermatchkov et al. for PZ carbamate and Bower et al. for DEA [2, p. 1509][3, Table 16][4, Table 5][5]. The AMP hydrolysis starting law is inferred from Xu et al.'s model-derived carbamate constants, and the DEA starting law is the inverse of Aroua et al.'s formation law [6, Table 2, Eqs. 7–11][7, Eq. 14]. Their fitted shifts are included in the completed coefficients; these are not independent measurements of the fitted reaction laws.

For a fitted enthalpy correction H about T_p, the correction added to the tabulated coefficients is

$\delta a = H/(RT_p)$ and $\delta b = -H/R$, preserving $\ln K(T_p)$. This correction is part of the reaction law and is not an independently added absorption-heat term.

Table S3
MDEA reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction
MDEA dissociation	−16.39891	−819.7	10.976	0	molality

Table S4
AMP reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction
AMP dissociation	−8.973099	−3850.736	8.064629	0	molality
carbamate hydrolysis	8.210491	−1944.838	0	0	molality

Table S5
PZ reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction
P1	−11.55919	−3123.533	6.786961	0	molality
P2	−2.804785	−2824.703	3.563322	0	molality
P3	8.791731	−3615.98	0	0	molality
P4	−10.04478	−3493.022	0	0	molality
P5	3.392648	−1322.144	0	0	molality

Table S6
DEA reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction
DEA dissociation	−9.279878	−4214.076	0	0.009961213	molality
carbamate hydrolysis	5.926332	−2631.648	0	0	molality

S2 Observation inventory and fitting objectives

No pressure set labeled held out enters any objective, including earlier fits shown for sensitivity. Prior scoring is nonetheless material: MDEA held-out sets were already inspected, AMP Kundu was available before final diameter choice, PZ Ermatchkov before the penalty alternative and final fit choice, and DEA holdouts before a numerically tied second-start choice. Comparison-only DEA properties were added after selection. A datum's absence from a final objective is not evidence of complete blindness during model development.

Table S7
Measurement locators preserved with the calculation inputs. Source row identities and original-observable columns accompany the electronic numerical files; fitted and unfitted roles are in the main paper.

Observation	Source	Locator
MDEA Kuranov	Kuranov1996 [8]	Table 2, printed p. 1960
MDEA Jakobsen	Jakobsen2005 [9]	Table A5, PDF p.9, printed p.9902
MDEA Carson	Carson2000 [10]	Table 3, PDF p.9, printed p.1293
MDEA Merkley	Merkley1987 [11]	Figure 3; PDF p.5 printed p.441
MDEA Cp	Weiland1997 [12]	Table 1, PDF p.1, printed p.1004
MDEA loaded-density	Weiland1998 [13]	Table 1, PDF p.2, printed p.379
MDEA unloaded-density	Sobrino2016 [14]	Table 5, PDF p.4, printed p.234
MDEA Kim	Kim2008 [15]	Table 8, PDF p.4, printed p.2524
AMP loaded total pressure	Hartono et al. (2021) [16]	Table 7; 47 fitted rows at 313, 333 and 353 K
AMP binary pressure and vapor composition	Hartono et al. (2013) [17]	Table 3; 26 pressure and 26 vapor-composition observations
AMP carbamate fraction	Ciftja et al. (2014) [18]	Supporting Information, Tables S1–S3; 57 observations
PZ fitted partial pressure	Hilliard (2008) [19]	Appendix D.5, printed pp. 916–917
PZ NMR groups	Ermatchkov et al. (2003) [4]	Tables 1–4; 121 states
PZ comparison-only total pressure	Kamps et al. (2003) [20]	Table 3, p. 2666; corrected temperatures: Ermatchkov (2006), Table 6, p. 1794; excludes precipitation row (313.75 K, 1.995 mol/kg, loading 1.69, 4.446 MPa)
PZ held-out partial pressure	Ermatchkov et al. (2006) [21]	Tables 1–3, printed p. 1789
DEA fitted partial pressure	Lawson and Garst (1976) [22]	Table II, PDF p. 4; 30 fitted rows
DEA fitted integral heat	Carson et al. (2000) [10]	Table 2; 50 rows at 298.15 K
DEA SeoHong	SeoHong1996 [23]	Table 1 (DEA/AMP = 30/0), PDF p.2, printed p.259
DEA Lal	Lal1985 [24]	Table 2, PDF p.2, printed p.682
DEA weiland	Weiland1998 [13]	Table 2, PDF p.2, printed p.379
DEA lawson viii	LawsonGarst1976 [22]	Table VIII (CO2 only; 77 F rows 54.5 wt% per footnote a), PDF p.10, printed p.29
DEA bottinger DEACOO	Bottinger2008 [25]	Table 4, PDF p.8, printed p.138
DEA cai	Cai1996 [26]	Table 4, PDF p.2, printed p.1102
DEA excess enthalpy	Horstmann2002 [27]	Table 6, PDF p.3, printed p.1498

Thus the dimensionless Kuranov scaled RMS is not its physical pressure AARD, and these group costs are not common physical accuracy measures. The stored rows preserve loading caps for integral heat, source amount bases for heat capacity and density, and unavailability of capped states.

The pressure, binary and species groups are not pooled when reporting their physical deviations. Kundu's 18, 25 and 30 mass-percent stocks correspond nominally to 2.0, 2.8 and 3.4 mol/L and span loading 0.412–0.966. Kundu's loading uncertainty is not a combined partial-pressure uncertainty, so it cannot supply an uncertainty-normalized pressure test.

The resulting 232 observations come from 121 source states and use a full block covariance.

Source group fractions are per total PZ, while source loading per twice the PZ amount is converted to the per-PZ convention before the bubble calculation.

DEA minimizes

$$C = \frac{1}{2}\left[\frac{1}{30}\sum_{j=1}^{30}\left(\frac{\ln(p_j^c/p_j^o)}{0.09}\right)^2 + \frac{1}{50}\sum_{j=1}^{50}\left(\frac{\Delta h_j^c - \Delta h_j^o}{0.02|\Delta h_j^o|}\right)^2\right]. \tag{S1}$$

Lawson pressures cover 310.93–380.37 K and loading no lower than 0.376 at a single fitted concentration. Carson heats cover mass fractions 0.10, 0.20 and 0.30 at 298.15 K.

S3 Starts, selected parameters and calibration residuals

Table S8
Original search bounds of the reported reactive fits. Reaction-level bounds refer to the fitted constant *a* before the enthalpy correction is added; DEA enthalpy bounds are refit increments. PZ values and ties follow the complete twelve-parameter fit. Bounds are not measured molecular intervals.

Amine	Parameter	Lower	Upper
MDEA	MDEA–water: k_{ij} at 313.15 K	−0.5	0.5
MDEA	MDEA–water: k_{ij} slope / K^{-1}	−0.01	0.01
MDEA	$MDEAH^+$–CO_2: k_{ij}	−0.5	0.5
MDEA	$MDEAH^+$–water: k_{ij}	−0.5	0.5
MDEA	$MDEAH^+$–water: k_{ij} reciprocal temperature slope	−1000	1000
MDEA	$MDEAH^+$–HCO_3^-: k_{ij}	−1	1
MDEA	$MDEAH^+$: segment diameter	3	4.5
MDEA	$MDEAH^+$: dispersion energy over k	50	950
MDEA	$MDEAH^+$: born diameter	2	7
AMP	AMP–water: k_{ij}	−0.5	0.5
AMP	AMP–water: k_{ij} reciprocal temperature slope	−1000	1000
AMP	$AMPH^+$–water: k_{ij}	−0.5	0.5
AMP	$AMPH^+$–water: k_{ij} reciprocal temperature slope	−1000	1000
AMP	$AMPH^+$–CO_2: k_{ij}	−1	1
AMP	$AMPH^+$: segment diameter	4.5	6.5
AMP	carbamate: segment diameter	5.32	6.5
AMP	carbamate hydrolysis a	8.555	13.555
AMP	carbamate hydrolysis enthalpy increment / J mol^{-1}	−5000	5000
DEA	$DEAH^+$–water: k_{ij}	−1	1
DEA	carbamate–water: k_{ij}	−1	1
DEA	carbamate hydrolysis a	4.56444	10.5644
DEA	carbamate hydrolysis enthalpy increment / J mol^{-1}	−19000	21000
PZ	Cation diameter / Å	2	9
PZ	Cation dispersion / K	50	1000
PZ	Cation–water interaction	−1	0.5
PZ	Cation–bicarbonate interaction	−1	0.5
PZ	Carbamate diameter / Å	2	9
PZ	Carbamate dispersion / K	50	1000
PZ	Carbamate–water interaction	−0.5	0.5
PZ	HPZCOO diameter / Å	2	6
PZ	HPZCOO dispersion / K	50	1000
PZ	P3, P4 and P5 offsets (each)	−3	3

Table S9
Retained start outcomes. Costs use different objective scales and must not be compared across amines. The MDEA A3 rows and smaller-diameter AMP fits are sensitivity comparisons, not additional selected results.

Fit / start	Cost	Iterations	Bounds	Termination
MDEA A3 fit	403.57153	10	4	converged
MDEA A3 alternative	403.57153	11	4	converged
MDEA fit	374.51908	11	2	converged
MDEA alternative	392.92418	17	2	converged
AMP reported floor	61.73030	45	2	converged
AMP 3.5 Å start 1	61.61537	54	2	converged
AMP 3.5 Å start 2	61.61537	52	2	converged
AMP 3.5 Å start 243	61.61537	28	2	converged
PZ reported	20698.620	100	1	iteration limit
DEA A-1	10.139543	8	0	converged
DEA A-2	10.139543	8	0	converged
DEA B-1	7.970700	6	0	converged
DEA B-2	7.970700	7	0	converged

The reported MDEA fit retains the fit with nine fitted parameters under fixed shared inputs. Its selected cost is 374.5190788 for 278 objective rows, with two active bounds; the second start converges to a larger cost, so agreement of two starts is not claimed. The A3 alternative is the same-data sensitivity counter-case, with no replacement of the reported MDEA fit. Its Carson scaled RMS increases from 0.97662 to 1.10594. Its selected parameter-file SHA-256 is `2a5341be176829d087133a1467828f762a655f92a586eb1062a5719a58142d9d`.

AMP uses the scored fit with floor 5.32 Å, cost 61.73030, rather than the lower-cost 3.5 Å alternative. All three starts of the latter reach the new smaller-diameter bound; the small cost change on restoring the larger floor shows weak identification below about 5.3 Å. The reported floor is the MEA carbamate diameter, while segment-volume scaling suggests a larger AMP carbamate, approximately 5.89 Å. Neither estimate is a measured ionic diameter. The completed hydrolysis enthalpy correction reaches −5000 J/mol in every refit of the sequence.

PZ's reported fit with twelve fitted parameters starts from a previous PZ fit mapped onto these ties. The independent proposed start is unavailable with HPZCOO included in the vapor; no successful fit from that start is implied. The reported fit reaches 100 iterations with cost 20698.6 nearly flat over its final fourteen iterations. A conditional diagnostic with the bound fitted parameter fixed converges at a cost lower by approximately 3×10^{-4}, but does not retroactively label the original termination converged. PZH^+–water reaches the lower bound -1. The separate penalty alternative is not substituted for this reported fit.

DEA compares two cation representations with two starts each. The two single-sphere starts give cost 10.13954, and the two charged-chain starts give 7.97070, all without active bounds. The charged-chain starts differ in cost by 4.8×10^{-10}, below the function tolerance, and their largest relative prediction difference is 2.5×10^{-6}. The selected second start therefore represents the same numerical optimum as the first to the stated resolution. Its held-out sets were first scored on the first start and then on this tied second start, whose parameters were fixed before the held-out sets were evaluated again. An unavailable more negative cation–water start is not counted as a converged solution.

The electronic data files contain all fitted pressure, heat and species predictions, including zeros and the actual observed-value denominators. PZ NMR error is reported in group fraction and whitened residual units; DEA NMR below is in true mole fraction. The same percentage formula does not make these different species observables comparable.

S4 Unfitted and comparison-only properties by source and range

The Ermatchkov MDEA total is a mixed-range summary, while every Xiao row is within the 278–368 K dissociation measurements. Kuranov's thirty extrapolated fitted rows are at 373.14–373.17 K and 393.12–393.15 K. The fitted Merkley heats give 10.36 % AARD at 333.15 K and 6.56 % at 388.71 K, each over two observations; signed errors are −5725 and −3989 J/mol CO_2, respectively. The latter is a dissociation-law extrapolation. The Carson, heat-capacity and loaded-density records lack the temperatures needed to group them by reaction-law range. Binary pressure and unloaded density do not depend on the reactive dissociation calculation.

AMP Kundu temperature blocks are 303, 313 and 323 K, with twenty-five, fifteen and eleven observations. Their pressure biases are +53.94, +43.78 and −7.09 %, respectively, whereas Tong's seventeen total-pressure observations span approximately 313, 333 and 353 K. Tong's twenty-three higher-temperature loaded rows are above the dissociation law's 353.22 K calculation limit and are not calculated. The source-law temperature basis of hydrolysis is extrapolated for the Kundu cold block even though the fitted hydrolysis law is evaluated there. These distinct law limits preclude one blanket in-range label for AMP.

PZ Ermatchkov has nineteen partial pressures at 313.1 K, eighteen at 353.15 K and fifteen at 393.15 K. Only the first block lies within the measured carbamate interval; the middle block already extrapolates carbamate. At the hottest temperature all fifteen calculated pressures are high. Eight observations at the lowest temperature, three at the middle and none at the highest lie within the printed uncertainty. The fifty-two pointwise source uncertainties are distinct from parameter uncertainty.

DEA Seo and Hong has five, six and five pressures at 40, 60 and 80 °C; Lal has seventeen and twenty-seven at 40 and 100 °C. Only the 40 °C blocks are within the Bower measured range. Lal states an accuracy of ±15 % above 0.05 kPa and much larger uncertainty below that pressure; five of twenty-eight higher-pressure rows fall within that stated interval. The sixteen lower-pressure observations remain in the overall AARD. At 100 °C, four very-low-loading values are described by Lal as probably in error (printed p. 684); they remain in the overall score.

DEA comparison-only Lawson Table VIII contains ten observations: two at 77 °F, four at 100 °F and four at 200 °F. The nominally fifty-mass-percent solution is 54.5 wt% in its coldest block. Cai's nine binary observations compare the model bubble pressure at measured liquid composition and temperature with 6.66 kPa. Weiland's forty-three loaded densities at 298.15 K exclude the documented misprinted twenty-mass-percent, loading-0.50 value. These comparisons did not enter the reactive objective or final parameter selection.

The Böttinger DEA comparison separates nine states without the oxazolidone byproduct from sixteen with byproduct. For the clean states, amine sum, carbamate and bicarbonate-sum mean absolute errors are 0.001749, 0.001301 and 0.001274 in true mole fraction. For byproduct-containing states the corresponding values are 0.003166, 0.004986 and 0.005347, and omitted byproduct chemistry prevents treating these as equivalent tests. The model uses assumed 2.5 MPa rather than unavailable individual cell pressures. No uncertainty for the NMR mole fractions is supplied, so these absolute errors are not uncertainty-normalized validation scores.

The fourteen Horstmann excess enthalpies are at 363.15 K and 1.75 MPa, with source scale $s = 2\ \mathrm{J/mol} + 0.01|H^E|$. The model gives positive mixing heat at every state while all observations are negative. The ideal-gas enthalpy contribution is linear in composition and cancels in excess enthalpy, which is calculated from the liquid residual enthalpies. The pure-water residual enthalpy is about −41.4 kJ/mol as a unit check. This is a comparison-only binary caloric result, separate from Carson's fitted absorption heats. No analogous assessment for the other amines establishes whether the error is specific to DEA.

Kamps total pressures are comparison-only, scored once after the fit with twelve fitted parameters was frozen. The temperatures use the correction in Ermatchkov et al., Table 6, p. 1794 [21]; the precipitation row at 313.75 K, 1.995 mol PZ/kg water, loading 1.69 and 4.446 MPa is excluded. The block comparison labels both PZ measurement intervals and the shared R2 computational interval; all 395.0 K rows exceed the latter. The two unloaded rows are included in the block and overall scores but separated from the

loaded pressure groups.

Table S10
Kamps total-pressure groups. Loaded high-temperature groups exclude unloaded rows and pressures below 0.2 MPa. Overall: 94 rows, 42.5 % AARD, −40.3 % bias.

Group	n	AARD / %	Bias / %
below 0.2 MPa	10	20.4	−6.0
374–395 K	35	37.1	−35.2
313–354 K, 0.2 MPa and above	47	53.0	−53.0
unloaded	2	1.0	0.1

Table S11
Kamps total-pressure blocks at corrected temperatures. Law labels: carbamate measurements 283.15–333.15 K; protonation measurements 293.14–353.42 K; shared R2 interval 293.15–393.15 K. Unloaded rows are included. Labels describe temperature ranges, not phase stability.

T / K	m_{PZ} / mol kg^{-1}	Carbamate	Protonation	R2	n	AARD / %	Bias / %
313.75	1.995	inside	inside	inside	7	32.1	−32.1
334.05	1.995	outside (+0.90 K)	inside	inside	8	35.6	−35.6
334.05	2.035	outside (+0.90 K)	inside	inside	6	48.2	−48.2
354.35	1.995	outside (+21.20 K)	outside (+0.93 K)	inside	13	33.4	−33.4
374.65	2.035	outside (+41.50 K)	outside (+21.23 K)	inside	10	29.4	−29.4
395.0	2.035	outside (+61.85 K)	outside (+41.58 K)	outside (+1.85 K)	10	23.7	−22.8
334.05	3.964	outside (+0.90 K)	inside	inside	13	76.3	−67.6
354.35	3.964	outside (+21.20 K)	outside (+0.93 K)	inside	8	56.5	−56.5
374.7	3.95	outside (+41.55 K)	outside (+21.28 K)	inside	10	49.8	−46.7
395.0	3.95	outside (+61.85 K)	outside (+41.58 K)	outside (+1.85 K)	9	32.4	−25.7

S5 Finite phase searches and unavailable states

Main-text Table 17 reports the finite state inventories rather than a phase diagram. For MDEA, distinct observation identities can represent repeated physical states; 775 records should not be described as 775 independent experiments. The thirty-seven gas-like phases are not dense-liquid splits, and all seven states above the 393.15 K R2 limit remain outside the phase evidence. Kuranov source rows 28 and 72 exceed shared R2's 393.15 K upper limit and are excluded from the 278-row fit.

The capped MDEA heat at 333.15 K, 60 wt% and 1.121 MPa could not be calculated.

AMP nominal Kundu states are searched at their calculated bubble states; loading-perturbed pressure evaluations do not supply additional phase evidence. Its calibration gas outcomes were not recorded and cannot be reconstructed as zero from the absence of a reported value. DEA's 140 searched fitted/held-out states comprise thirty pressures, fifty heats, sixteen Seo and Hong and forty-four Lal states. The additional comparison-only sets are not included in that count. No additional liquid or gas phase was detected in those DEA searches, and no global-stability conclusion follows.

PZ searches 121 NMR liquid states and sixty-two Hilliard bubble states. Where an assumed 0.1 MPa NMR liquid gives a gas-like split, it is moved to 1.05 times its calculated bubble pressure before the liquid check. The reported fit has negative tangent-plane distances at Hilliard rows 55, 56, 61 and 62 of approximately -0.096, -0.111, -0.0024 and $-0.031\ RT$. These are false liquid splits at the observed liquid compositions, not experimental phase-separation measurements. Solid equilibrium is absent, including at temperatures below the pure-PZ melting region.

At Kamps bubble states the search detects thirty-six dense-liquid splits, finds no additional phase at fifty-six states, and leaves two unloaded states undecided; no gas or ambiguous split is found. The thirty-six detected splits are a lower bound; the fifty-six states without detected splitting are not proven stable, and the two unloaded states remain undecided. At 1.5 times bubble pressure the standard finite search finds splits at Hilliard rows 55 and 62 but misses rows 56 and 61. Direct tangent-plane distances at the known split compositions are -0.0957, -0.1107, -0.0024 and $-0.0306\ RT$ at rows 55, 56, 61 and 62, respectively. All 121 NMR states pass at $T+0.5$ K and 0.1 MPa, except that three rich states are evaluated at 1.05 times bubble pressure where it exceeds 0.1 MPa. These checks do not establish a continuous stable region.

Table S12
Kamps phase census at 94 scored bubble states. Split counts are lower bounds; states without a detected split do not prove stability.

Outcome	States
No additional phase found	56
Resolved dense-liquid split	36
Undecided (unloaded rows)	2
Gas split	0
Ambiguous	0

The separate penalty fit uses the Part I tangent-plane requirement at 245 states: the 121 NMR liquids and each Hilliard feed as an all-liquid parent at 1.05 and three times its candidate bubble pressure. Penalty scales are 10^{-2} followed by 3×10^{-3}. Its standard finite search reports one split, but direct distance calculations at split compositions recovered near the fit show two further negative values at rows 55 and 56. Those direct values, $-0.0309\ RT$ and $-0.0437\ RT$, demonstrate that the finite search missed resolved splits. The penalty did not establish a single-liquid fit. Consequently detected-split counts are lower bounds in both Part I and this paper, and a count of states without detected instability is an upper bound on stability within the searched states, not proof of global stability.

S6 Conditional covariance, sensitivity and bound profiles

All parameter errors are local and conditional on the species model, fixed shared inputs, data weights and active bounds. They exclude experimental systematic errors, future measurement noise, model discrepancy and uncertainty in transferred constants. Covariance propagation includes parameter correlations, while the maximum one-SE sensitivity in the tables moves one parameter marginally with the others fixed. For a bound parameter, the last column instead uses the first inward profile step with other parameters refitted. Those perturbation sizes are different; their common display is not a statistical significance ranking. Parameter values and standard errors use Å for diameters, K for dispersion energies divided by k_B and reciprocal-temperature interaction slopes, K^{-1} for linear-temperature interaction slopes, and J mol^{-1} for hydrolysis enthalpy shifts; interaction constants, hydrolysis a coefficients and P3–P5 offsets are dimensionless.

Table S13
MDEA conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
$MDEAH^+$: ϵ/k_B	623.04	36.686	0.87587
$MDEAH^+$, water: k_{ij}	−0.033236	0.0099899	0.66959
$MDEAH^+$: σ	3.6494	0.030615	0.096832
$MDEAH^+$, water: water interaction slope	−27.105	3.3628	0.090821
MDEA–water a0: coefficient	−0.20634	0.0013412	0.044204
MDEA–water a1: coefficient	0.001475	2.5631×10^{-5}	0.041912
$MDEAH^+$, CO_2: k_{ij}	0.24819	0.029345	0.0046476
$MDEAH^+$, HCO_3^-: k_{ij}	1	–	0.0042142
$MDEAH^+$: born_diameter	7	–	0.0013609

Table S14
AMP conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
$AMPH^+$: σ	5.4153	0.43273	0.62404
$AMPH^+$, water: k_{ij}	−0.31364	0.038238	0.52636
$AMPH^+$, water: water interaction slope	−274.15	85.929	0.12653
AMP, water: k_{ij}	−0.13074	0.0060466	0.11814
AMP, water: water interaction slope	−55.961	15.867	0.033161
$AMPH^+$, CO_2: k_{ij}	0.75499	0.087274	0.030051
carbamate hydrolysis a	8.2105	0.29086	0.020545
carbamate: σ	5.32	–	0.00039001
carbamate hydrolysis δh / J mol^{-1}	−5000	–	3.5166×10^{-5}

Table S15
PZ conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
PZH^+/u	102.73	21.234	1.6031
HPZCOO/u	975.43	113.63	1.0564
carbamate/u	717.59	141.82	0.89897
HPZCOO/sigma	3.8143	0.066159	0.80996
PZH^+/bicarbonate-k	−0.021035	1.031	0.78658
PZH^+/sigma	4.2856	0.32334	0.51166
carbamate/sigma	4.8625	0.35603	0.50409
P4/offset	−0.019326	0.13257	0.47204
carbamate/water-k	0.07407	0.049728	0.3681
P3/offset	0.15704	0.12251	0.35292
P5/offset	−0.26155	0.060658	0.058636
PZH^+/water-k	−1	–	0.0031208

Table S16
DEA conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
carbamate hydrolysis a	5.9263	0.5973	0.57223
carbamate, water: k_{ij}	0.0057148	0.052851	0.46078
$DEAH^+$, water: k_{ij}	−0.21516	0.0034456	0.083032
carbamate hydrolysis δh / J mol^{-1}	−12216	1309.3	0.071383

MDEA has two active bounds and conditional variance factor 2.7639784. The inward cation–bicarbonate interaction profiles give cost increases 0.142, 0.7282 and 1.508 at values 0.98, 0.90 and 0.80. Born-diameter profiles at 6.95, 6.75 and 6.50 Å give increases 0.06927, 0.3571 and 0.7433. All converge, but this limited grid does not locate a confidence limit or a global minimum. The two weakest full-Jacobian directions include both bound parameters, with singular values 0.00848001 and 0.0198683 in the original optimizer units. Their inclusion of constrained fitted parameters limits interpretation of two-sided displacements.

AMP has two bound fitted parameters and variance factor 0.829. The $AMPH^+$ diameter and cation–water interaction have correlation about 0.96; cation–water interaction and slope have correlation about 0.98. Inward carbamate diameter steps of 1, 5 and 10 % of the original interval give cost changes 0.0009, 0.0044 and 0.0087. The corresponding enthalpy-correction steps give changes 0.0063, 0.032 and 0.064. The twelve retained AMP profile fits, including the interior cation–CO_2 interaction diagnostic, converge. Nearly flat bound profiles restrict molecular interpretation even when covariance bands are comparatively small.

PZ holds one bound fitted parameter fixed and uses eleven interior fitted parameters with variance factor 146.3. The conditional refit moves enough that the inverse Jacobian at the refitted parameters is used rather than identifying its covariance with the original fit without qualification. The 1 % inward profile converges; 5 and 10 % profiles stop at their forty-iteration budget, with cost changes 46.5 and 61.9. Those two changes are upper bounds on minimized profile cost. The weakest full twelve-parameter Jacobian singular direction has singular value about 0.21 per optimizer unit and is dominated by cation–bicarbonate and cation–water interactions. Large marginal pressure sensitivities make its linear estimates indicative rather than reliable finite-displacement predictions.

DEA has no active bounds and uses variance factor 0.210. Its hydrolysis-law level and carbamate–water interaction have correlation 0.972, so an individual marginal sensitivity can exceed the correlated prediction error. The source-separated pressure SEs are much smaller than the held-out errors, and fitted heat SEs are only 0.9–1.9 %. This is conditional information within the chosen model; it does not apportion the remaining error between experiments and thermodynamic formulation.

S7 AMP substitution sequence and source-comparison qualifications

The complete AMP sequence uses the same fifty-one Kundu observations at every step, with no Kundu residual entering any objective. The earlier shared inputs give 43.69 % Kundu AARD and +40.51 % bias. Substituting Part I's shared bicarbonate–water interaction and R2 without refitting gives 21.89 % and −15.10 %, while worsening the fitted total pressures. Refitting reverses that apparent improvement, as shown in Table S17. This is a numerical attribution under the specified substitutions, not an identified physical defect or proof that either experimental source is wrong.

Table S17
AMP same-data sequence. Calibration uses total pressure and carbamate fraction; Kundu uses partial pressure and remains outside every objective. The fourth row is the reported fit.

Change	Cost	Fitted P AARD / %	Species RMS	Kundu AARD / %	Bias / %
Shared substitution; no AMP refit	238.38	18.85	0.01386	21.89	−15.10
Original AMP limits	66.28	9.93	0.00614	43.51	40.35
Carbamate diameter floor raised	66.42	9.93	0.00637	43.03	39.77
Cation–CO2 interaction widened	61.73	10.33	0.00631	42.85	37.79
Carbamate diameter floor lowered	61.62	10.34	0.00586	43.95	39.09

The earlier bicarbonate–water interaction is 0.72705584775, compared with one in Part I. The earlier R2 coefficients $a = 23.58551614210$ and $b = -11928.05673746$ K change to $a = 23.87092013090$ and $b = -12017.43099656$ K, while $c = -36.7816$ remains fixed in the same molality basis. The six common component descriptions do not otherwise change in this sequence. The comparisons are same-data parameter substitutions, not an independent physical-validation experiment.

A separate data-only comparison of Hartono and Kundu lacks sufficient overlap to decide whether the systematic Kundu error reflects experiment or the model. There is no adequate 303 K overlap, and the exploratory interpolation requires concentration and vapor-pressure-baseline assumptions. Its exploratory ratios are therefore not used as independent quantitative evidence in this paper. The source-versus-model question for the 42.85 % Kundu error remains open.

S8 Numerical identities and display provenance

The four studies use the merged ePC-SAFT source commit `391dfe5d64ede5036bf160bc6ae037ad5e14af32` and immutable wheel SHA-256 `0962e21d2fa7a8ba1ed4622f1110b3e8053bacd878fa41c999d85e8b9afab66b`. The common MEA parameter input has SHA-256 `f8f64820b161377a0d97d1e4ad4f6cea764cd0558dc63231d58e2096988eef97`. The electronic input index binds the exact amine-specific files and laws to those shared inputs. Recalculation checks verify numerical reproduction and balance; they are not physical validation.

MDEA recalculation from the parameter file reproduces all 278 fitted predictions with zero stored prediction difference and relative cost difference 2.76×10^{-13}. Recalculation from the AMP parameter file reproduces the scored cost within 7.4×10^{-13} relative. DEA recalculation from the parameter file reproduces cost within 8×10^{-13} relative and weighted residuals within 1.3×10^{-11}. PZ retains its iteration-limited fit, conditional diagnostic, residuals and all phase trials, including the direct split witnesses missed in the penalty fit's search. Numerical precision does not account for the larger experimental discrepancies.

The presentation script reads JSON and CSV values without importing the thermodynamic model.

The accompanying numerical-source index maps table cells and manuscript paragraph values to source files and fields; figure values and parameter-only bands remain separate from experimental uncertainties. All plots use discrete observations and model evaluations at the measured states; no interpolated model sweep or new phase boundary is introduced.

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